Renyi Yang

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Education

ShanghaiTech University — Shanghai, China
B.Eng. in Computer Science and Technology — *Sep. 2023 – Present*
- GPA: 3.84/4.0; Major GPA: 4.0/4.0
- Class rank: 10/170 in CS; 16/265 in CS + EE

Grenoble INP — Grenoble, France
Signal Image Processing Methods and Applications — *Jan. 2026 – Jun. 2026*

Internships

miHoYo (Hoyoverse)
AI Engineer Intern — *Jul. 2026 – Present*
- Built **LLM pipelines** for warehouse and shop-revenue reconciliation and sales analytics
- Developed **social sentiment early-warning** and **KOL** benchmarking for sponsorship decisions
- Produced **per-version competitor analyses** and **social-media campaign insights** to guide strategy

Shanghai Azimuth Data Technology Co., Ltd.
Software Engineer Intern — *Jul. 2025 – Sep. 2025*
- Developed a **multi-COM** GNSS client that concurrently parses **NMEA 0183** and **RTCM 3.3** streams
- Built a **LAN-based** simulator client to both ingest and transmit messages

Industry Collaborations

Lynred
Data Challenge 2026 — *May 2026 – Jun. 2026*
- Detected and corrected **vertical stripe defects** in 16-bit infrared image sequences across sensors
- **Ranked first** among all competing teams

AMD
Open Hardware 2025 — *Sep. 2025 – Jan. 2026*
- Developed **HeteroOCR**, a CPU/GPU/NPU-optimized OCR system for complex documents
- Offloaded OCR to the NPU: **40%** lower GPU utilization, **2**× faster than CPU
- Showcased at **CIIE 2025** as AMD's flagship NPU-AI demo

Research Experience

Post-Moore Microelectronics and Integrated Circuit Center (PMICC)
Research Assistant, advised by Prof. Yajun Ha — *Jul. 2025 – Present*
- Developing a **high-throughput Chinese OCR** pipeline for heterogeneous case files
  - Layout, Detection, Recognition; Watermark Removal; Acceleration

Projects

Network on Acoustic Link | *Go, full network stack* [GitHub] — *Sep. 2025 – Dec. 2025*
- Implemented physical, MAC, network, transport and application layers over an acoustic link

Oat Compiler | *OCaml, LLVM* [optimizations] [type-checking] — *Sep. 2025 – Dec. 2025*
- Built a compiler pipeline: lexing and parsing → LLVM → assembly, with type checking and optimization

Pintos | *C, x86 operating system* [GitHub] — *Mar. 2025 – Jun. 2025*
- Implemented threads, user programs and syscalls, virtual memory, and an extensible file system

Plants vs. Zombies | *C++, CMake* [GitHub] — *May 2024 – Jun. 2024*
- Built a desktop game using a GameObject base-class architecture

Awards & Honors

AMD Open Hardware 2025 Competition [results] — HeteroOCR
136 Hours Volunteer Service — China Volunteer Service
3+1 Overseas Exchange Program Scholarship — ShanghaiTech University
Outstanding Student — ShanghaiTech University

Technical Skills

Languages: Go, C&C++, OCaml, Python, MATLAB, RISC-V, X86lite
Tools & Frameworks: Git, Linux, Docker, Conda, Vim, GNU Make, CMake
Other: LaTeX, Markdown, Pi, DSH, CC-Switch